

which are Memstore and Hfile. Regions are mainly tables that are split up and spread across the region servers.

Region server: When a region server receives writes and read requests from the client, it assigns the request to a specific region, where the actual column family resides. However, the client can directly make contact with region servers — there is no need of mandatory master permission to the client for communication with region servers. The client requires master help when operations related to metadata and schema changes are required.

Why use HBase?

Today, every Web application consists of billions of rows. Searching for a few particular rows from a large amount of data takes a lot of time. In such a situation, HBase is the ideal choice as the query fetch time is short. Conventional relational data models fail to meet the performance requirements of very big databases.

HBase vs HDFS vs Hive

HBase	HDFS	Hive
HBase is built on top of HDFS.	HDFS is a distributed file system suitable for storing large files.	It is a relational DBMS.
HBase provides fast look-ups for larger tables.	HDFS does not support fast individual record look-ups.	Hive does not provide fast look-ups.
It provides low latency access to single rows from huge datasets.	It provides high latency batch processing.	It provides high latency for huge datasets.
Provides random access to data.	Provides sequential access to data.	Provides random access to data.

HBase vs RDBMS

HBase	RDBMS
Schemaless in database.	Has fixed schema in database.
Column-oriented database.	Row-oriented database.
Designed to store de-normalised data.	Designed to store normalised data.
Well suited for OLAP systems.	Well suited for OLTP systems.
It is good for semi-structured as well as structured data.	It is good for structured data.

HBase's limitations

1. HBase cannot execute functions like SQL. It doesn't support the SQL structure, so it does not contain any query optimiser.
2. We cannot expect to completely use HBase as an alternative for conventional models, some of which cannot hold HBase.
3. HBase, when integrated with Pig and Hive jobs, results in some time memory issues in the cluster.

You can download HBase from the Apache website, the latest version of which is 1.2.4. The HBase team recommends that you install it on a UNIX/Linux environment; if you run it in Windows, you might want to download and install Cygwin to do so. 

By: Neetesh Mehrotra

The author works at TCS as a systems engineer. His areas of interest are Java development and automation testing. You can contact him at mehrotra.neetesh@gmail.com.

OSFY Magazine Attractions During 2017-18

MONTH	THEME
March 2017	Open Source Firewall, Network security and Monitoring
April 2017	Databases management and Optimisation
May 2017	Open Source Programming (Languages and tools)
June 2017	Open Source and IoT
July 2017	Mobile App Development and Optimisation
August 2017	Docker and Containers
September 2017	Web and desktop app Development
October 2017	Artificial Intelligence, Deep learning and Machine Learning
November 2017	Open Source on Windows
December 2017	BigData, Hadoop, PaaS, SaaS, IaaS and Cloud
January 2018	Data Security, Storage and Backup
February 2018	Best in the world of Open Source (Tools and Services)